

the required components are listed in the Bill of Materials table.

Prerequisite

The Arduino IDE is used to write the software, and the ESP32 board must be installed on the IDE. Follow the steps provided on the ESP website to

add the ESP32 to the Arduino IDE. After installation, the libraries for the DHT sensor, ACS712 current sensor, and LCD display must be added via the Arduino Library manager.

MQTT and Adafruit IO are employed to display the data on the IoT dashboard. Sign in to an

Adafruit IO account and create an API by following the steps outlined by Adafruit IO. For more information, visit: <https://io.adafruit.com/api/docs/#about-the-api-docs>. Additional details can be found here: <https://cdn-learn.adafruit.com/downloads/pdf/adafruit-io.pdf>.

Software

In the source code, begin by defining the pins for the sensors and LCD display, then include the necessary libraries. Configure the Wi-Fi SSID and password for the

```
Adafruit_MQTT_Publish Feed4 = Adafruit_MQTT_Publish(&mqtt, AIO_USERNAME "/feeds/level");
Adafruit_MQTT_Publish Feed5 = Adafruit_MQTT_Publish(&mqtt, AIO_USERNAME "/feeds/state");

void MQTT_connect();

#include <LiquidCrystal.h>
LiquidCrystal lcd( D_05, D_18, D_19, D_21, D_22, D_23);

#include <OneWire.h>
#include <DallasTemperature.h>
#define ONE_WIRE_BUS 25
float Celsius = 0;
OneWire oneWire(ONE_WIRE_BUS);
DallasTemperature sensors(&oneWire);

const int trigPin = 4;
const int echoPin = 2;
long duration;
double level, distanceCm;
float TankHeight = 20;
```

Fig. 3: Setting pins for LCD and sensors in the code



Fig. 5: Local data monitoring on the LCD

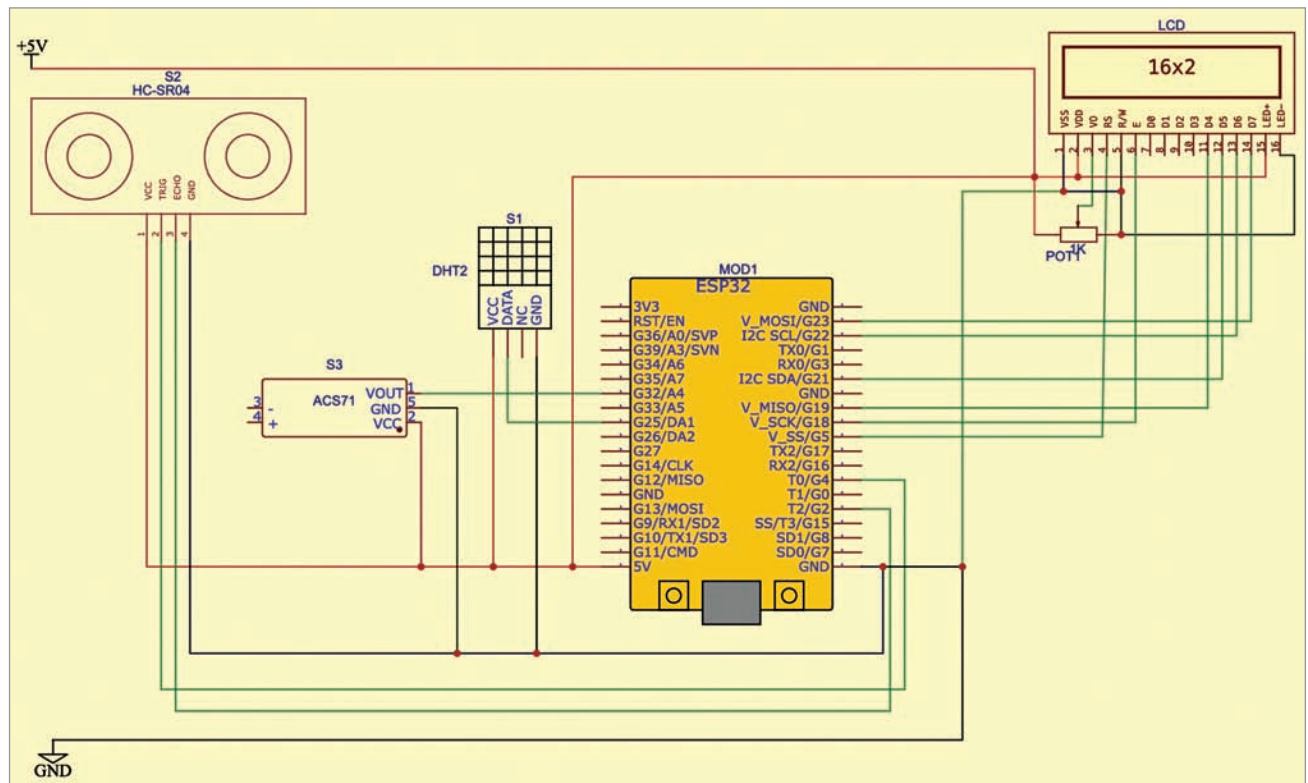


Fig. 4: Circuit diagram